King Fahd University of Petroleum and Minerals

Department of Mathematics

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Math 101 Major Exam I Term 251

Original exam date: not printed. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

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1. $\lim_{x\to-\infty}\left(x+\sqrt{x^2+3}\right)=$

(a) DNE

(b) ∞

(c) 3

(d) $-\infty$

(e) 0

2. If $f(x)=\begin{cases}x^2\sin\dfrac{1}{x}, & x\neq 0\\ 0, & x=0\end{cases}$ Then

(a) f is not differentiable at $x=0$

(b) $f'(0)=2$

(c) $f'(0)=0$

(d) $f'(0)=\dfrac{1}{2}$

(e) $f'(0)=1$

3. $\lim_{x\to1}\dfrac{|x+1|-|-x-1|}{x-1}=$

(a) 2

(b) 0

(c) -2

(d) DNE

(e) -1

4. The value of a for which the function

$$f(x) = \begin{cases} \dfrac{x^2 - a^2}{x - a}, & x \neq a \\ 8, & x = a \end{cases}$$

is continuous is equal to

(a) -2

(b) 0

(c) 2

(d) 4

(e) 8

5. $\lim_{x \to 2} (2x - [\![x]\!]) =$

(a) 0

(b) 2

(c) 3

(d) 1

(e) DNE

6. The value of k for which the line $y = \dfrac{kx}{4} + 4$ is tangent to the graph of $f(x) = k\sqrt{x}$ is equal to

(a) 4

(b) -2

(c) 2

(d) $\dfrac{1}{4}$

(e) -4

7. $\lim_{x \to 0} \dfrac{2\cos x - 2}{x^2} =$

(a) $\dfrac{1}{2}$

(b) DNE

(c) 1

(d) 0

(e) -1

8. The graph of the function $f(x) = \dfrac{4x^2 + 4x - 24}{x^4 - 2x^3 - 9x^2 + 18x}$ has

(a) Three vertical asymptotes and one hole

(b) One vertical asymptote and one hole

(c) Two vertical asymptotes and two holes

(d) No vertical asymptotes and four holes

(e) Four vertical asymptotes and no holes

9. The graph of the function $f(x) = \dfrac{|x(x+1)|}{x^2 - 1}$ has

(a) One horizontal and one vertical asymptotes.

(b) No horizontal and one vertical asymptotes.

(c) Two horizontal and two vertical asymptotes.

(d) One horizontal and two vertical asymptotes.

(e) Two horizontal and one vertical asymptotes.

10. If the points $P_1(2, 0)$, $P_2(0, y_1)$, and $P_3(1, y_2)$ lie on the graph of $y = f(x)$ and the slope of the secant line through P_1 and P_2 is 2 while the slope of the secant line through P_1 and P_3 is 3, then $f(0) + f(1) =$

(a) -3

(b) -7

(c) 21

(d) 1

(e) -1

11. $\lim_{x \to 2^-} \dfrac{\dfrac{x+1}{x} - \dfrac{3}{2}}{x - 2} =$

(a) -1

(b) $\dfrac{1}{2}$

(c) $-\infty$

(d) $-\dfrac{1}{4}$

(e) ∞

12. $\dfrac{d}{dx}\left(e^2 + x^e + \dfrac{2}{\sqrt[3]{x}} + 3\cos x\right) =$

(a) $ex^{e-1} - \dfrac{2}{3x^{1/3}} + 3\sin x$

(b) $ex^{e-1} + \dfrac{2}{3x^{4/3}} - 3\sin x$

(c) $ex^{e-1} - \dfrac{2}{3x^{4/3}} - 3\sin x$

(d) $ex^{e} \ln x - \dfrac{2}{3x^{4/3}} - 3\cos x$

(e) $2e + ex^{e-1} + 2x^{-\frac{2}{3}} - 3\sin x$

13. If $f(x) = \begin{cases} x^2 + 1, & x \leq 2 \\ 4x - 2, & x > 2 \end{cases}$ Then

(a) $f'(2) = 4$

(b) f is not differentiable at $x = 2$

(c) $f'(2) = \dfrac{1}{2}$

(d) $f'(2) = 1$

(e) $f'(2) = 2$

14. $\lim_{x\to 0} \dfrac{1 - e^{-x} + 4\left(e^{2x} - 1\right)}{e^x - 1} =$

(a) DNE

(b) $\dfrac{7}{9}$

(c) 7

(d) 9

(e) $\dfrac{9}{7}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	Conjugate: $\frac{-3}{x-\sqrt{x^2+3}} \to 0$.
2	**(c)**	$f'(0) = \lim x \sin \frac{1}{x} = 0$ by squeeze.
3	**(b)**	$\lvert -x-1\rvert = \lvert x+1\rvert$, numerator is 0, so the limit is 0.
4	**(d)**	$\lim_{x\to a}(x+a) = 2a = 8$, so $a = 4$.
5	**(e)**	Right limit 2, left limit 3, so DNE.
6	**(a)**	Slopes equal give $x = 4$; then $2k = k+4$, so $k = 4$.
7	**(e)**	Multiply by $\cos x + 1$: $-2(\frac{\sin x}{x})^2 \frac{1}{\cos x+1} \to -1$.
8	**(c)**	$f = \frac{4}{x(x-3)}$: asymptotes $x = 0, 3$; holes $x = -3, 2$.
9	**(a)**	$f = \frac{x}{x-1}$ for large $\lvert x\rvert$: only $x = 1$ vertical, only $y = 1$ horizontal.
10	**(b)**	$y_1 = -4$, $y_2 = -3$, so $f(0) + f(1) = -7$.
11	**(d)**	Combine: $\frac{2-x}{2x(x-2)} = -\frac{1}{2x} \to -\frac{1}{4}$.
12	**(c)**	$0 + ex^{e-1} - \frac{2}{3}x^{-4/3} - 3\sin x$.
13	**(b)**	Jump at $x = 2$ (5 vs 6), so not continuous, not differentiable.
14	**(d)**	$\frac{1-e^{-x}}{e^x-1} = e^{-x} \to 1$; $4(e^x+1) \to 8$; sum 9.